

Ethical AI Storyteller

Integrating Safety, Bias Mitigation, and Explainability in Neural Story Generation

Eron Salihu Miran Mustafa Urim Hoxha

Faculty of Electrical and Computer Engineering
University of Prishtina “Hasan Prishtina”

Natural Language Processing Course
Instructor: Dr. Sc. Mërgim H. HOTI

December 2025

Outline

- 1 Introduction
- 2 System Architecture
- 3 Datasets
- 4 Methodology
- 5 Implementation
- 6 Evaluation
- 7 Discussion
- 8 Conclusion

The Challenge

Large language models can generate fluent narratives but face critical challenges:

- **Safety:** Risk of producing toxic or harmful content
- **Bias:** Amplification of societal stereotypes
- **Transparency:** Black-box decision making

Our Solution

An integrated system that combines story generation with:

- Multi-layer ethical filtering
- Pattern-based bias detection
- Explainability mechanisms

Project Contributions

① Modular Storyteller

- GPT-2 generation with layered ethical filtering
- Real-time bias detection

② Built-in Explainability

- Attention heatmaps
- Token importance analysis
- Next-token diagnostics

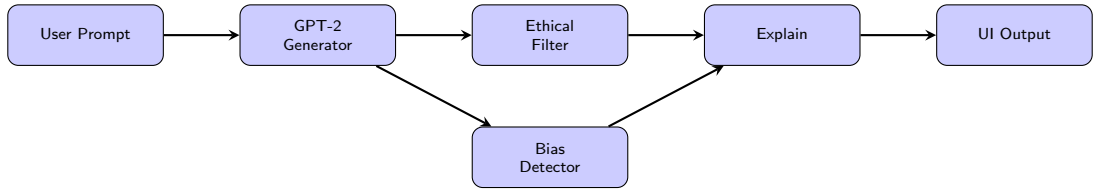
③ Reproducible Pipeline

- Fine-tuning support
- Offline evaluation
- 50-story local dataset

④ Interactive Interface

- Real-time Gradio web application
- Safety ratings and bias alerts

System Overview



Core Modules

Generation & Processing

- `story_generator.py`
 - GPT-2 models
 - 8 genre support
 - Nucleus sampling
- `dataset.py`
 - HuggingFace datasets
 - Local JSON stories
- `finetune.py`
 - Training pipeline
 - Checkpointing

Ethics & Transparency

- `ethical_filter.py`
 - Detoxify integration
 - Pattern matching
 - 5-level rating
- `explainability.py`
 - Attention visualization
 - Token importance
- `storyteller.py`
 - Main orchestrator
 - Retry logic

Datasets Overview

Dataset	Source	Samples	Size
WritingPrompts	HuggingFace	500–1000	~50MB
TinyStories	HuggingFace	500–1000	~30MB
FairyTales	HuggingFace	200–500	~15MB
Local Dataset	Custom	50	45KB

Local Dataset Characteristics

- **Purpose:** Offline reproducibility for grading
- **Genres:** Fantasy (20%), Sci-Fi (18%), Mystery (16%), Adventure (16%), Romance (16%), Horror (14%)
- **Format:** JSON with id, genre, prompt, story

Story Generation

Model Variants

- GPT-2 (124M) – default for efficiency
- GPT-2 Medium (355M)
- GPT-2 Large (774M)
- DistilGPT-2 (82M) – fastest inference

Generation Parameters

- Nucleus sampling ($p = 0.95$)
- Top- $k = 50$
- Temperature = 0.8–1.0
- Repetition penalty = 1.2
- 3-gram blocking
- Max length = 200 tokens

Multi-Layer Ethical Filtering

Layer 1: ML-Based Toxicity Detection

Detoxify with thresholds: toxicity 0.5, severe 0.3

- Detects: hate speech, threats, identity attacks, obscenity

Layer 2: Pattern-Based Blocking

Regex patterns for explicit harmful content

- Violence, discrimination, self-harm keywords

Layer 3: Prosocial Encouragement

Detects positive elements: friendship, hope, courage, kindness

Content Rating System

$$\text{rating}(y) = f_{\text{safety}}(y; \tau) \in \left\{ \begin{array}{l} \text{Safe} \\ \text{Mild} \\ \text{Moderate} \\ \text{Restricted} \\ \text{Blocked} \end{array} \right\}$$

- **Blocked patterns** → BLOCKED
- **High toxicity** (>0.8) → BLOCKED
- **Moderate toxicity** (0.6–0.8) → RESTRICTED
- **Sensitive topics** → MODERATE/MILD
- **Positive content** (3+ cues) → SAFE

Retry Mechanism

Re-generates (max 3 attempts) with increased temperature if unacceptable.

Pattern-Based Approach

Identifies stereotypes in generated text:

Category	Example Patterns
Gender roles	"women should...", "men must..."
Profession	"woman nurse", "man engineer"
Trait	"emotional woman", "tough man"

Actionable Recommendations

When bias detected:

- Specific category identified
- Suggestions for balanced representations
- Warning in UI

1. Attention Visualization

Heatmaps showing token attention (selectable layer)

2. Leave-One-Out Token Importance

Measures contribution via KL divergence:

$$\text{importance}(t_i) = D_{KL}(P_{\text{baseline}} \| P_{\text{masked}})$$

3. Next-Token Prediction

Top-k candidates with probability distribution

Web Interface (Gradio)

Interactive Features

Tab 1: Generate Story

- Genre selection, parameter tuning, safety ratings

Tab 2: Continue Story

- Extend existing narratives with coherence

Tab 3: Explain

- Next-token predictions, attention heatmaps, token importance

Tab 4: Safety Check

- Analyze any text with detailed toxicity scores

Offline Mode & Reproducibility

Designed for Coursework

- **Offline operation:** Works without internet
- **Local dataset:** 50-story JSON bundle
- **Graceful fallback:** Regex-only if Detoxify unavailable
- **Model caching:** Pre-download for grading

Testing Strategy

- **Unit tests:** Mocked modules (no large model loading)
- **Smoke tests:** Dataset loading, offline init
- **Benchmarks:** Optional execution

Perplexity (PPL)

Measures model confidence (lower is better):

$$\text{PPL} = \exp \left(-\frac{1}{N} \sum_{i=1}^N \log P(\text{token}_i | \text{context}) \right)$$

Safe Content Ratio

Proportion passing safety filters: $\text{Safe Ratio} = \text{safe_count} / \text{total_count}$

Bias Ratio

Proportion with detected stereotypes: $\text{Bias Ratio} = \text{biased_count} / \text{total_count}$

Results: Model Comparison

Model	PPL ↓	Safe ↑	Bias ↓
GPT-2 base	41.12	100%	0%
GPT-2 (fine-tuned)	35.30	100%	0%

Table: Evaluation on 50-sample local dataset

Key Findings

- **14% perplexity improvement** after fine-tuning on WritingPrompts
- **100% safe content** with regex-only filtering
- **0% bias detection** on clean evaluation set

Ablation Study Results

Setting	Unsafe Caught	Bias Caught	Total Issues
No filter	0	0	0
Regex-only	1	0	1
+ Bias checks	1	3	4
+ Detoxify	4	3	7

Table: Ablation on 27 test samples (19 edge cases + 8 generated)

Insights

- Each layer catches different types of issues
- Detoxify catches **3 additional cases** missed by regex
- Bias detector identifies **3 stereotypical patterns**
- Full pipeline: 21/27 (78%) clean outputs

Throughput Performance

Model	CPU (tokens/s)	MPS GPU (tokens/s)
GPT-2	79	62
DistilGPT-2	115	76

Table: Apple Silicon M-series performance

Computational Complexity

- **Filtering:** Linear in text length
- **Generation:** Scales with sequence length & model size
- **Attention extraction:** One forward pass
- **Leave-one-out:** Multiple passes (expensive, best for demos)

Qualitative Example

Generated Story (Fantasy)

Prompt: “In a magical kingdom, a young wizard...”

Output: “...they chose kindness over revenge and helped the village rebuild...”

Safety: SAFE (0.85) **Bias:** None **Positive:** kindness, helped

Generate–Inspect–Revise Workflow

- 1 Generate story with parameters
- 2 View safety/bias analysis
- 3 Adjust and regenerate as needed
- 4 Experiment with safety awareness

Limitations & Future Work

Current Limitations

- Rule-based bias detection, English-focused
- Modest dataset (50 local stories)
- Incomplete coverage for nuanced harms

Future Directions

- Semantic bias detection
- Multilingual support
- RLHF-style safety tuning
- Larger datasets, controllable generation
- Quantization for low-resource devices

Ethical Considerations

Safety

Multi-layer filtering, revisions suggested, retry mechanism

Bias Awareness

Pattern detection, recommendations, conscious choices

Transparency

Explainability, inspect reasoning, trust through interpretability

Note: Human review for sensitive deployments

Conclusion

Key Achievements

- Safety, bias detection, explainability integration
- 14% perplexity improvement
- Multi-layer filtering
- Interactive Gradio interface

Demonstration

Ethical constraints integrated without sacrificing fluency

Educational Impact

Foundation for responsible AI and NLP ethics

Repository Structure

- `src/` – Core modules
- `app.py` – Gradio interface
- `data/sample_stories.json` – Local dataset
- `tests/` – Unit tests
- `paper/` – IEEE paper

Quick Start

`python app.py` – Launch interface

`python finetune_hf.py --dataset tiny_stories` – Fine-tune

`python src/evaluate.py --model gpt2` – Evaluate

Thank You!

Questions?

Ethical AI Storyteller

NLP 2025 – Group 3
University of Prishtina